SYSTEMATIC TRADING · BACKTEST REVIEW

Swing Strategy Backtest

Trend Pullback — rules, Indian costs, and the evidence

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SIMULATED

The strategy finished the test period at **₹546,660** on ₹500,000 of starting capital, a +9.3% total return (+1.0% a year) against an equal-weight buy-and-hold of the same universe at +105.8%.

+1.0%	-5.1%	261	+0.04R
CAGR after costs	max drawdown	round trips	average trade

TEST WINDOW	2017-01-02 to 2025-09-30 (8.7 years)	**STARTING CAPITAL**	₹500,000
UNIVERSE	6 symbols, daily bars	**COSTS**	₹33,223 paid
DATA	Synthetic simulation — not market prices	**STATUS**	Backtest · no orders were placed

> **The data in this report is simulated**
>
> Every price in this run comes from tools/make_sample_data.py, a random-walk generator. The results describe how this strategy behaves against a plausible-looking series, and nothing at all about real NSE or BSE securities. Replace data/*.csv with real daily bars and rerun before treating any figure here as evidence.

WHAT THIS IS	One fully specified long-only swing strategy, run across a fixed universe with Indian delivery costs charged per leg, and judged on what it did rather than what it promised.
WHAT IT IS NOT	Not a signal service, not a forecast, and not connected to any broker. No order was placed to produce it. A backtest is a record of the past dressed in a suit.
THE ONE NUMBER TO LOOK AT	Average R — profit per trade as a multiple of what was risked. It already includes costs, so it is the closest thing here to an edge.

1. The verdict, in one line

Not yet an edge

Positive, but thin: a profit factor of 1.20 and +1.0% a year would not survive a bad quarter, a slippage assumption or a tax bill. Treat this as a hypothesis, not a strategy.

The rest of this document does three things: states the rules as code (section 2), shows what they did and what it cost (sections 3 to 7), and then argues against itself (section 8). Read section 8 before deciding the earlier pages mean anything.

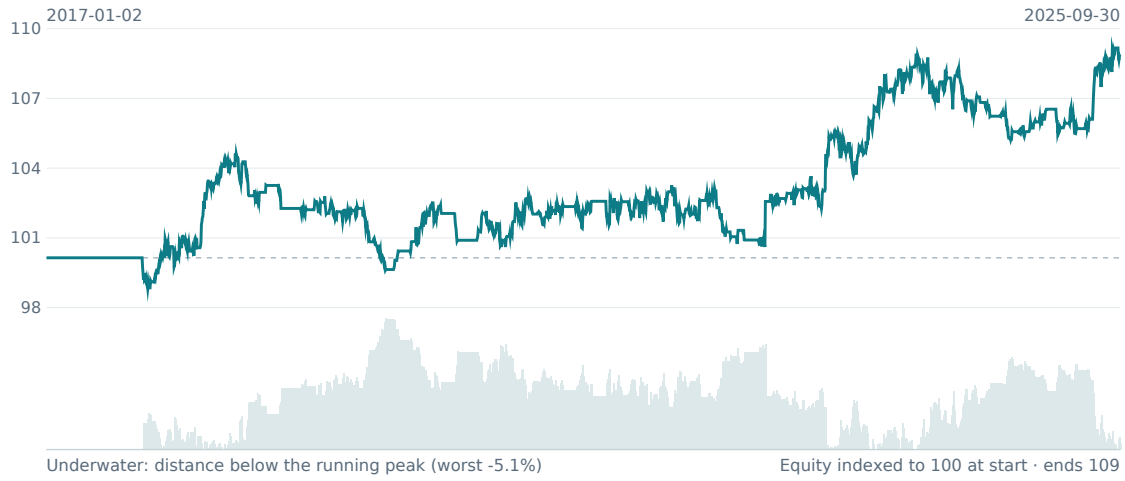
2. The rules under test

Every rule below is fixed in code before the first trade. Nothing is discretionary, which is the only way a backtest can mean anything: if you would have overridden the system in March, you did not test the system you would have traded.

ELEMENT	RULE	WHY IT IS THERE
Regime filter	close > SMA200 and SMA50 > SMA200	Only buy pullbacks inside an uptrend. Most losing swing trades are pullbacks in downtrends.
Entry trigger	RSI(2) ≤ 12	A short, sharp washout — not a slow bleed — is what a healthy pullback looks like.
Fill	next bar's open	Removes lookahead. A signal you cannot act on is not a signal.
Stop	entry − 2.5 × ATR(14)	Volatility-scaled: a wide, quiet stock gets room; a jumpy one gets a tighter leash.
Target	entry + 4 × ATR(14)	A fixed, pre-committed exit. The reward:risk ratio is known before entry.
Exits	stop, target, RSI(2) ≥ 80, close < SMA200, or 40 bars	A trade that goes nowhere costs opportunity and attention; time is an exit.
Sizing	risk 1.0% of equity per trade, capped at 20% of equity	Size follows the stop distance, not conviction. This is what keeps ruin off the table.
Portfolio	4 concurrent positions, one per symbol	Correlated entries in a falling market is one big position wearing four hats.

3. Equity curve and drawdown

The upper line is account equity, indexed to 100 at the start. The shaded band is how far the account sat below its own high-water mark. A curve you cannot look at during a drawdown is a curve you will abandon at the worst possible moment.



4. Results, after Indian delivery costs

Charges are modelled trade by trade — STT, exchange, SEBI, stamp duty, GST and DP — and total ₹33,223 across the test, which is 6.6% of the starting capital. A gross-profit figure without this deduction is fiction.

Total return after costs	**+9.33%**	Expectancy per trade	**₹179**
CAGR	**+1.03%**	Average trade	**+0.04R**
Max drawdown	**-5.10%**	Average win / loss	**₹1,796 / ₹-2,342**
Sharpe ratio (daily, annualised)	**+0.37**	Average holding period	**5.9 bars**
Round trips	**261**	Total costs paid	**₹33,223**
Win rate	**60.9%**	Time in the market	**16.8% of capacity**
Profit factor	**1.2**	Equal-weight buy and hold	**+105.81%**

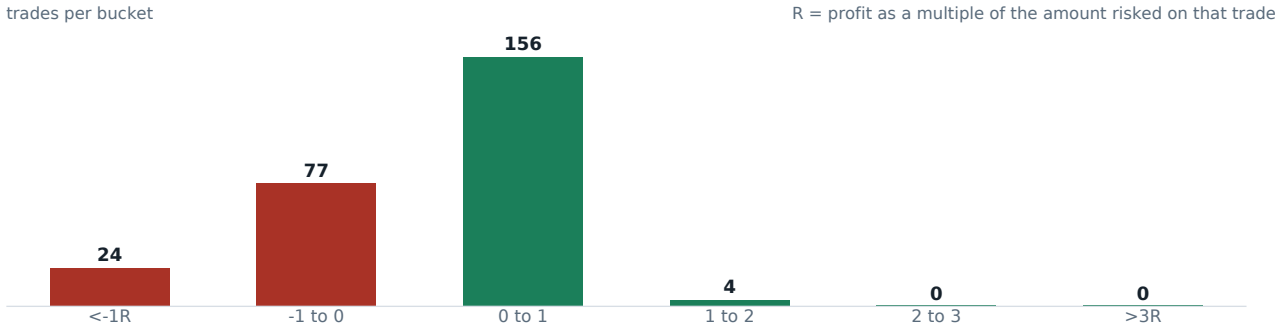
5. Cost model applied

Rates as charged on NSE delivery trades in 2025, applied per leg inside the backtest rather than as a flat haircut at the end.

CHARGE	RATE	SIDE
Securities transaction tax	0.1%	buy and sell
Exchange transaction charge (NSE)	0.00325%	buy and sell
SEBI turnover fee	0.0001%	buy and sell
Stamp duty	0.015%	buy only
GST on brokerage + charges	18%	buy and sell
DP charge (per sell, per scrip)	₹15.93	sell only
Brokerage (discount broker, delivery)	₹0	buy and sell

6. Shape of the trades

A strategy is a distribution, not an average. What matters is whether the right tail is fat enough to pay for the cluster of small losses on the left. Win rate on its own is close to meaningless.



7. Where the money came from

Per-symbol results tell you whether one lucky name carried the whole test. If a single symbol is responsible for most of the profit, you have a one-stock bet with extra steps.

SYMBOL	TRADES	WIN RATE	NET P&L	SHARE OF TRADES
ALPHA	55	60.0%	**₹6,724**	21%
BRAVO	51	62.7%	**₹10,168**	20%
CHARLIE	56	58.9%	**₹7,021**	21%
DELTA	18	55.6%	**₹-8,179**	7%
ECHO	55	70.9%	**₹33,744**	21%
FOXTROT	26	46.2%	**₹-2,817**	10%

How positions were closed

EXIT	COUNT	AVG R	NET P&L
Overbought	194	+0.24	**₹237,464**
Trend break	39	-0.39	**₹-78,462**
Stop	24	-1.02	**₹-126,982**
Target	2	+1.58	**₹16,007**
End of test	2	-0.12	**₹-1,367**

The worst trades — by design, these exist

A stop is not a guarantee. The loss column shows what each one actually cost once costs and gaps were included.

SYMBOL	ENTRY	EXIT	BARS	REASON	NET P&L
CHARLIE	2024-02-15	2024-02-20	4	stop	**₹-5,562**
CHARLIE	2023-12-18	2023-12-21	4	stop	**₹-5,525**
CHARLIE	2024-06-12	2024-06-19	6	stop	**₹-5,502**
ALPHA	2024-07-12	2024-07-22	7	stop	**₹-5,487**
CHARLIE	2024-03-13	2024-04-01	14	stop	**₹-5,473**
BRAVO	2025-03-24	2025-03-27	4	stop	**₹-5,473**

8. What this test cannot tell you

- **The data is not real.** Synthetic prices have no earnings, no circuits, no news, no illiquidity and no fat-tailed panic. Real markets are ruder than any generator, so treat every figure here as an upper bound on how the rules behave.
- **Slippage is assumed away.** Fills are at exact open, stop and target prices. In practice a stop in a fast market fills worse, and thin stocks move against you on entry. Budget 0.1–0.3% per side and rerun before believing the result.
- **The universe was chosen with hindsight.** These six symbols exist because they were available, and a survivor is by definition a stock that survived. A test on today's index members is quietly a test of winners.
- **Circuits and halts are missing.** Indian equities have daily price bands. A stock locked at the lower circuit cannot be exited at any price, which turns a 1% risk into a much larger one.
- **Income tax is not in these numbers.** The backtest deducts transaction charges but not tax on the gains. Held under twelve months, profits are short-term capital gains taxed at 20%; 261 round trips in 8.7 years also means 261 rows to report. A strategy must clear that hurdle too.
- **Overfitting is easy and invisible.** Six parameters were chosen by a human who could see the data. That is already a form of fitting. Walk-forward testing — parameters tuned on 2017–2021, judged on 2022–2025 — is the minimum defence.

9. What would earn this strategy real money

01 **Replace the data**
Export real daily bars for your actual universe from your broker or NSE bhavcopy, drop them in data/ and rerun. Nothing above matters until this is done.

02 **Walk forward**
Tune nothing; simply run the same fixed rules across 2017–2019, 2020–2021 and 2022–2025 separately. Consistency across periods is the only credible signal.

03 **Add friction**
Charge 0.15% slippage per side and rerun. If the edge disappears, it was never an edge — it was a rounding error.

04 **Paper trade for 30 signals**
Log every signal the system produces, and every one you were tempted to skip. Compare the two lists at the end. That gap is you.

05 **Trade the smallest size that is real**
Only after the four steps above, and only with an amount whose total loss would change nothing in your life.

Not investment advice. This document is a software artefact: a record of what a fixed set of rules did on a dataset, produced by tools/trade_engine.py. It is not a recommendation, an offer, or a solicitation to buy or sell any security, and it does not account for your circumstances. No order was placed to produce it, and no broker connection exists in this repository. Anyone who tells you a backtest is a forecast is selling something.

Artefacts: **backtest-trades.csv** (every round trip, with R-multiples) · **backtest-equity.csv** (daily equity) · **backtest-metrics.json** (all figures above, machine-readable).